

Logarithmic Green Domains at Actual Whitney Cross-Caps

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Abstract

An arclength coordinate along the degenerate direction of a smooth Euclidean Whitney germ makes its scalar divergence tensor equal to the identity up to $O(\rho^{1/2})$. Subtracting the planar logarithmic fundamental solution therefore leaves a divergence source in L^3 and a scalar source in L^2 . The scalar Hölder theorem gives a continuous regular part. For the fixed-Dirichlet two-point restriction on the signal normalization this identifies the actual resolvent Green vectors, proves their universal logarithmic coefficient, and gives geometric boundary coordinates on the whole adjoint domain. These coordinates establish Markov uniqueness and invariance under a common change of logarithmic reference length, as well as sheet-even limiting coefficients. The abstract extension framework is classical. The result repairs the scalar domain statements for a specified operator; singular Dirac and Pin transmission and publication priority remain separate questions.

1 The operator and the comparison being made

The source is the completed physical rectangle of the map

$$S(t, u) = (c + 2u(1 - c), (1 - u)s, u\sqrt{c(2 - c)}), \quad c = \cos t, \quad s = \sin t,$$

with $-\pi/2 \leq t \leq \pi/2$ and $0 \leq u \leq 1$. Endpoints use the completed smooth charts of the v13 manuscript [1]. Let P_+, P_- be its two interior Whitney rank-loss points, let $g = S^*g_{\mathbb{R}^3}$, and let $\mathcal{H} = L^2(dA_g)$. Our scalar product is conjugate-linear in its first argument. The closed gradient form with fixed outer Dirichlet condition defines the positive self-adjoint operator H_F . The previous continuation [2] establishes graph-bounded, surjective values

$$\tau : \text{Dom } H_F \longrightarrow \mathbb{C}^2, \quad \tau u = (u(P_+), u(P_-)), \quad \overline{\ker \tau}^{\mathcal{H}} = \mathcal{H}.$$

Write $A = H_F|_{\ker \tau}$. It is closed, dense and symmetric, with deficiency indices $(2, 2)$. For $\eta < 0$ put

$$R_\eta = (H_F - \eta)^{-1}, \quad G_\eta = (\tau R_\eta)^*, \quad G_{\eta,j} = G_\eta e_j.$$

The written Hilbert-space argument of [2], also an instance of Posilicano's established framework [4], gives

$$\text{Dom } A^* = \text{Dom } H_F \dot{+} G_\eta \mathbb{C}^2, \quad A^*(u + G_\eta a) = H_F u + \eta G_\eta a. \quad (1)$$

The purpose here is to turn these resolvent coordinates into actual asymptotic coefficients. The local argument concerns full neighborhoods of the two interior cross-caps. No extra boundary coefficients are introduced at the other three rank-loss labels: their Friedrichs form and the outer boundary condition are already fixed.

This choice matters. The graph closure H_0 of the compactly supported interior Laplacian leaves infinitely many outer-boundary modes free. Its deficiency and harmonic kernel are infinite-dimensional [1]. Statements below about A cannot be read as a finite classification of H_0 .

2 An asymptotically Euclidean coordinate

At an actual smooth Euclidean Whitney germ, a target translation and rotation, followed by a smooth source coordinate change, put the map in the form

$$\begin{aligned} F(x, y) &= (x, h(x, y)), & h(0) &= 0, & Dh(0) &= 0, \\ h_y(x, y) &= xv + 2yw + O(R^2), & R &= (x^2 + y^2)^{1/2}, \end{aligned} \quad (2)$$

where h takes values in $\mathbb{R}^2$ and v, w are linearly independent. Indeed, take x to be the target tangent-axis projection of F , and choose y along its kernel at zero. The Whitney determinant $\det(F_x, F_{xy}, F_{yy})(0) \neq 0$ gives the independence of v, w . This uses the actual Euclidean metric; an arbitrary target diffeomorphism would not preserve it. In a sufficiently small neighborhood, $cR \leq |h_y| \leq CR$ off zero and $|h_x| \leq CR$.

Lemma 2.1 (Arclength resolution). *Set*

$$V(x, y) = \int_0^y |h_y(x, s)| ds, \quad \Xi(x, y) = (X, V) = (x, V(x, y)), \quad \rho = (X^2 + V^2)^{1/2}.$$

The map Ξ is a local homeomorphism and a C^1 diffeomorphism off zero. In its coordinates the metric matrix G , its density $m = \sqrt{\det G}$, and its divergence tensor $a = mG^{-1}$ satisfy

$$G - I = O(\rho^{1/2}), \quad a - I = O(\rho^{1/2}), \quad m - 1 = O(\rho). \quad (3)$$

They extend continuously at zero with $G(0) = a(0) = I$, $m(0) = 1$, and are bounded and uniformly elliptic near zero. The actual extrinsic radius $\rho_e = |F(x, y)|$ satisfies

$$\rho_e/\rho = 1 + O(\rho^{1/2}), \quad \log \rho_e - \log \rho = O(\rho^{1/2}). \quad (4)$$

Proof. For fixed x , V is strictly increasing in y . The bounds on h_y imply

$$cy^2/2 \leq |V(x, y)| \leq C(|xy| + y^2).$$

Thus its inverse is continuous at zero and elsewhere, and its image contains a neighborhood of zero. Put $n = h_y/|h_y|$ away from zero. Differentiation under the integral, justified by the uniform bound on h_{xy} and dominated convergence, gives

$$V_y = |h_y|, \quad V_x = \int_0^y n(x, s) \cdot h_{xy}(x, s) ds, \quad |V_x| \leq C|y|.$$

The single undefined integrand value at $(x, s) = (0, 0)$ has no effect. These derivatives are continuous, including at zero. Since $V_y > 0$ off zero, the inverse-function theorem gives the asserted C^1 regularity. We need no second derivative of V along $x = 0$.

Let $q = h_x - V_x n$. Differentiating the actual map in the new coordinates yields $F_X = (1, q)$, $F_V = (0, n)$, and hence

$$G = \begin{pmatrix} 1 + |q|^2 & q \cdot n \\ q \cdot n & 1 \end{pmatrix}, \quad \det G = 1 + |q|^2 - (q \cdot n)^2 \geq 1.$$

Here $|q| \leq CR$. Since $y^2 \leq C|V|$, we have $R^2 \leq C\rho$ after shrinking the neighborhood. This proves (3) and uniform ellipticity, including the better density estimate.

To compare radii, Taylor's formula gives

$$|h - wy^2| \leq C\{x^2 + |xy| + |y|R^2\}, \quad |V - \operatorname{sgn}(y)|w|y^2| \leq C\{|xy| + |y|R^2\}.$$

The second estimate follows by integrating $||h_y(x, s)| - 2|w||s|| \leq C(|x| + x^2 + s^2)$. It follows that $||h| - |V|| \leq CR(|x| + y^2) \leq CR\rho$. Comparing the norms of $(x, |h|)$ and $(x, |V|)$ proves $|\rho_e - \rho| \leq CR\rho$. As $R \leq C\rho^{1/2}$, both radius claims follow. $\square$

Grieser already proved singular quasi-Euclidean coordinate resolution for the standard Whitney metric [3]. The present proof supplies the asymptotic identity tensor and the extrinsic-radius normalization for an actual Euclidean germ. This is a refinement to be compared with the literature, not a claim to have originated singular coordinate resolution.

3 The actual Green vectors

We use the following external scalar regularity input [5]: if a is real, bounded and uniformly elliptic in dimension two, an H^1 weak solution of $-\operatorname{div}(a\nabla w) = f_0 - \operatorname{div} F_0$ is locally Hölder when $F_0 \in L^q$, $f_0 \in L^{q/2}$, $q > 2$. Bounded zeroth-order terms can be moved to f_0 ; here $q = 3$ and $w \in L^2$. Real and imaginary parts are treated separately. Simon's Lecture 18, Theorems 2 and 3, printed pages 212–216, give the local boundedness and Hölder estimates with precisely these hypotheses.

Theorem 3.1 (Logarithmic normalization). *At P_i , with $\rho_{e,i} = |S - S(P_i)|$, there are real constants $B_{\eta,ij}$ and $\beta > 0$ such that*

$$G_{\eta,j} = -\frac{\delta_{ij}}{2\pi} \log \rho_{e,i} + B_{\eta,ij} + O(\rho_{e,i}^\beta). \quad (5)$$

The matrix B_η is real symmetric. The Green vector has the fixed outer Dirichlet condition inherited from R_η .

Proof. Work first with real functions at one pole in the coordinate of Lemma 2.1. The form is locally $\int \nabla f \cdot a \nabla v \, dX \, dV$ and the measure is $m \, dX \, dV$. Take a cutoff $\chi = 1$ near zero, supported in this chart, and set $h_0 = -\chi \log \rho / (2\pi)$. It is in $\mathcal{H}$, but not in the form domain. Distributionally $-\Delta h_0 = \delta_0 + k$, where k is supported in the cutoff annulus and is bounded. Write $E = a - I$. The remaining functional in

$$(-\operatorname{div}(a\nabla) - \eta m)h_0 = \delta_0 + \mathcal{F}$$

is represented by $E\nabla h_0$ in the divergence term, together with k and $-\eta m h_0$ in the scalar term. In particular

$$|E\nabla h_0| \leq C\rho^{-1/2}, \quad E\nabla h_0 \in L^3, \quad k - \eta m h_0 \in L^2.$$

Indeed $\int_0^\epsilon \rho^{1-3/2} d\rho < \infty$. The cutoff annulus adds only bounded terms. Extending by zero makes $\mathcal{F}$ a bounded functional on the global form domain. Lax–Milgram for the coercive form $q_\eta = q - \eta\langle \cdot, \cdot \rangle$ gives a unique form-domain w with $q_\eta(w, v) = \mathcal{F}(v)$. The quoted scalar theorem applies to its weak equation (including the $\eta m w$ term in L^2), so w is Hölder at the pole. Thus $g = h_0 - w$ has a logarithm and a Hölder regular part, and solves the point-source equation. Away from its pole it is a weak homogeneous solution and is Hölder at the other pole as well. Equation (4) replaces $\log \rho$ by $\log \rho_e$, reducing the exponent, if necessary, to $\beta \leq 1/2$.

We must still identify this solution with the resolvent-defined vector; distributional existence by itself would not suffice. For $u \in \operatorname{Dom} H_F$, scalar regularity gives $|u - u(P_i)| \leq C\rho^\alpha$. Testing its weak equation with a cutoff times $u - u(P_i)$ gives the local Caccioppoli bound

$$\int_{B_r} |\nabla u|^2 \leq Cr^{2\alpha}, \quad 0 < r < r_0,$$

after taking $0 < \alpha < 1$. The gradient-cutoff term is bounded by $Cr^{-2} \int_{B_{2r}} |u - u(P_i)|^2 \leq Cr^{2\alpha}$; the scalar source term is at most $Cr^{1+\alpha} \|m H_F u\|_{L^2(B_{2r})}$, which has the same bound for small r . Ellipticity absorbs the cross term.

The improper integral $\int a \nabla h_0 \cdot \nabla u$ therefore converges: on dyadic annuli $\|\nabla \log \rho\|_{L^2}$ is bounded, while $\|\nabla u\|_{L^2} = O(r^\alpha)$, a summable bound. Let θ_ϵ vanish on B_ϵ and equal one off $B_{2\epsilon}$, with $|\nabla \theta_\epsilon| \leq C/\epsilon$. The form test $h_0 \theta_\epsilon$ in the equation for u has an extra term bounded by $C|\log \epsilon| \epsilon^\alpha \rightarrow 0$. Consequently

$$\langle h_0, (H_F - \eta)u \rangle = u(P_i) + \mathcal{F}(u).$$

For the identity on the right, the planar radial flux of $-\log \rho / (2\pi)$ has total mass one; replacing u by its circle mean gives $u(P_i)$ in the limit, with error $O(\epsilon^\alpha)$. The E and scalar terms converge

in their form-dual pairings. The equation for w similarly gives $\langle w, (H_F - \eta)u \rangle = \mathcal{F}(u)$. Thus $\langle g, (H_F - \eta)u \rangle = u(P_i)$ for every $u \in \text{Dom } H_F$. Taking $u = R_\eta f$ identifies $g = (\tau_i R_\eta)^* 1 = G_{\eta,i}$. Complex functions follow by linearity with the stated scalar-product convention.

Reality of the form and η makes all $G_{\eta,j}$ and their regular constants real. For reciprocity, let $\delta_{\epsilon,i}$ be the normalized indicator of a resolved ball about P_i , normalized using dA_g . Graph Hölder estimates imply $\langle \delta_{\epsilon,i}, \cdot \rangle \rightarrow \tau_i$ in the dual graph norm, so $R_\eta \delta_{\epsilon,i} \rightarrow G_{\eta,i}$ in $\mathcal{H}$. Near a different pole these functions solve the homogeneous equation; the local Hölder and boundedness estimates improve convergence to uniform convergence on a smaller ball. Self-adjointness of R_η gives

$$\langle \delta_{\epsilon,i}, R_\eta \delta_{\epsilon,j} \rangle = \langle \delta_{\epsilon,j}, R_\eta \delta_{\epsilon,i} \rangle.$$

For $i \neq j$, passage to the limit gives $G_{\eta,j}(P_i) = G_{\eta,i}(P_j)$. The diagonal entries are already real, so B_η is real symmetric. $\square$

4 Geometric coordinates on the whole adjoint domain

Theorem 4.1 (Expansion and Green pairing). *Every $f \in \text{Dom } A^*$ has unique coefficients $\ell, b \in \mathbb{C}^2$ with*

$$f = \ell_i \log \rho_{e,i} + b_i + O(\rho_{e,i}^\beta) \quad \text{at } P_i. \quad (6)$$

For $f = u + G_\eta a$ in (1), they are

$$\ell = -a/(2\pi), \quad b = \tau u + B_\eta a.$$

Both coefficient maps are bounded in the graph norm of A^* , and the combined map is onto $\mathbb{C}^2 \oplus \mathbb{C}^2$, with kernel $\text{Dom } A$. For f, g with coefficients (ℓ, b) and (ℓ', b') ,

$$\langle A^* f, g \rangle - \langle f, A^* g \rangle = 2\pi \{ \langle \ell, b' \rangle - \langle b, \ell' \rangle \}. \quad (7)$$

The Friedrichs domain is precisely $\ell = 0$. All self-adjoint extensions are the inverse images of maximal isotropic planes for (7). Equivalently they have, for a unique $U \in U(2)$,

$$(U - I)b + i(U + I)a = 0, \quad a = -2\pi\ell.$$

Here $U = I$ gives H_F . This parametrization differs by the regular-part shift B_η from the resolvent coordinates in [2].

Proof. Theorem 3.1, the graph Hölder estimate for u , and (1) give the expansion. Its uniqueness follows by first dividing by $\log \rho_e$ and then taking the remaining limit. Moreover $u = R_\eta(A^* - \eta)f$ is graph-bounded, and the finite-dimensional inverse on $G_\eta \mathbb{C}^2$ bounds a . Surjectivity follows by prescribing a and then using surjectivity of τ to prescribe $b - B_\eta a$. Their common kernel is $a = 0$, $\tau u = 0$, namely $\text{Dom } A$.

The actual resolvent decomposition gives the pairing $\langle \tau u, a' \rangle - \langle a, \tau u' \rangle$. Substituting $\tau u = b - B_\eta a$ cancels the B_η terms by symmetry and gives (7). Vanishing a is exactly $\text{Dom } H_F$ in the direct sum. The finite-dimensional maximal-isotropic classification now applies to this surjective actual boundary map; its Cayley description gives the displayed $U(2)$ equation. $\square$

For $z \in \rho(H_F)$, retain the previous Weyl matrix $M_z = (z - \eta)\tau R_z G_\eta$. The actual geometric regular-part matrix is

$$B_z = B_\eta + M_z, \quad (8)$$

because $G_z - G_\eta = (z - \eta)R_z G_\eta \in \text{Dom } H_F$. If a self-adjoint matrix C specifies $b = Ca$, the resolvent is

$$(H_C - z)^{-1} = R_z + G_z(C - B_z)^{-1}G_z^*,$$

whenever the inverse exists. The Krein extension of A has plane $b = B_0 a$ and kernel $G_0 \mathbb{C}^2$. These are consequences of the existing extension framework, with a now-proved interpretation of its coefficients.

Corollary 4.2 (Actual reflection symmetry). *For real $\eta < 0$ the matrix B_η has the form $\begin{pmatrix} r_\eta & g_\eta \\ g_\eta & r_\eta \end{pmatrix}$. The same holds at $z = 0$.*

Proof. The actual map satisfies $S(-t, u) = JS(t, u)$, where $J(X, Y, Z) = (X, -Y, Z)$ is Euclidean orthogonal. Source reflection preserves the induced form, area and fixed outer Dirichlet condition, so its unitary pullback commutes with H_F and R_η and exchanges $G_{\eta,+}$ and $G_{\eta,-}$. It exchanges the extrinsic radii exactly. The two diagonal constants are consequently equal; reciprocity supplies the equal real off-diagonal constants. The argument also applies to R_0 since H_F has a positive gap. This symmetry exchanges the two poles; it is distinct from the local sheet exchange at a single cross-cap. $\square$

5 Two uniqueness statements and local parity

Corollary 5.1 (Markov uniqueness for the specified restriction). *H_F is the only self-adjoint extension of A whose heat semigroup is submarkovian.*

Proof. The closed gradient form with outer Dirichlet condition is a Dirichlet form: normal contractions decrease the gradient energy and preserve its closure. Thus its operator H_F generates a submarkovian semigroup. Conversely, for any such self-adjoint extension H , $\lambda > 0$, and $h \in L^2 \cap L^\infty$, the Laplace-transform formula for its resolvent gives

$$\|(H + \lambda)^{-1}h\|_\infty \leq \lambda^{-1}\|h\|_\infty.$$

By (6), any nonzero logarithmic coefficient makes this resolvent vector essentially unbounded near its pole; the regular remainder is bounded and the resolved measure has a positive bounded density. Hence every coefficient vanishes. The vector lies in $\text{Dom } H_F$, and $(H + \lambda)^{-1}h = (H_F + \lambda)^{-1}h$. Bounded L^2 inputs are dense, so the resolvents, and therefore the self-adjoint operators, are equal. $\square$

Corollary 5.2 (Common reference-length invariance). *Among self-adjoint boundary planes for A , the only one invariant under all simultaneous changes of logarithmic reference length is the Friedrichs plane.*

Proof. Writing $f = \ell_i \log(\rho_{e,i}/\lambda_i) + b_i^{(\lambda)} + o(1)$ gives $b^{(\lambda)} = b + \text{diag}(\log \lambda_i)\ell$. The change $\lambda_i \mapsto e^t \lambda_i$ acts by $(\ell, b) \mapsto (\ell, b + t\ell)$. If a maximal isotropic plane L is invariant for every real t , then $(\ell, b) \in L$ implies $(0, \ell) \in L$. Pairing these vectors in (7) yields $2\pi\|\ell\|^2 = 0$, so $\ell = 0$. Dimension two gives the entire Friedrichs plane. It is itself invariant. $\square$

This statement changes a reference convention. It does not assert a dilation isometry of the actual surface or a physical renormalization law.

Corollary 5.3 (Sheet-even limiting coefficients). *If paired source points p_ϵ, q_ϵ approach the same interior cross-cap and $S(p_\epsilon) = S(q_\epsilon)$, then every $f \in \text{Dom } A^*$ satisfies $f(p_\epsilon) - f(q_\epsilon) = O(\rho_\epsilon^\beta) \rightarrow 0$. Thus the logarithmic and regular boundary coefficients are sheet-even.*

Proof. The common image has exactly the same extrinsic radius on both branches. Their logarithmic terms in (6) cancel, their constant terms coincide, and both remainders tend to zero with the stated bound. $\square$

This is asymptotic parity of scalar coefficients. It requires no sheet isometry of the actual metric and asserts no invariance of the full Green function. The ordinary support obstruction and the odd-to-even equivariant linear obstruction of v13 remain applicable. No canonical derived-sheaf-to-operator morphism, phase-fiber map or singular Pin transmission is supplied.

6 What has been repaired, and what is still missing

The ancestral TeX and the authoritative final v12 are different records. The 1,908-line ancestor supplies the earlier signal motivation; the 43-page final v12 adds the late scalar and spinorial assertions. This continuation proves logarithmic coefficients and scalar uniqueness for A , replacing the ill-defined finite classification of the old H_0 . It recovers the intended two-point conclusion under a specified fixed outer boundary, not under the old compact-core definition.

The construction in continuation 0.02 uses component powers $(1 - u, u)$. The ancestral moment assignment was $(\cos t, 1 - \cos t)$. They are different state families. The explicit new quadrature realization proves that S can be realized with supplied phases; it does not derive S from the ancestral moment map alone. The Hermitian no-go also explains why changing this readout into three pure-state quantum expectations is impossible.

Abstract boundary triples and resolvent formulas are established theory [4]. Markov uniqueness for degenerate operators also has a substantial earlier literature [6]; its theorem for a compact core on an open Euclidean domain is not imported here, because our outer boundary is fixed and its coefficient hypotheses differ. The potential contribution lies in the actual Euclidean Whitney estimates, coefficient normalization and the fully specified application. Independent derivation and a new proof in this project do not establish publication priority. That question needs further comparison with singular-metric and point-interaction literature.

The principal mathematical tasks still open are the actual singular Dirac graph domains, conformal and gauge estimates, front/seam overlap control, and a proved phase-fiber/Pin transmission map. The scalar Hölder theorem does not prove a Dirac system estimate. The earlier weight-zero commutator problem and the missing fiber identification remain unresolved. Physical clock, spin, chirality, channel capacity or arithmetic realizations need their own objects and assumptions. The new scalar PDE and operator proofs are written proofs using external theorems; the existing 33-theorem Lean inventory has not been enlarged or reinterpreted as full formalization.

References

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